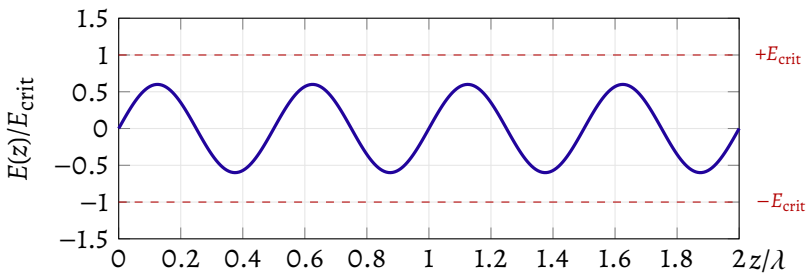


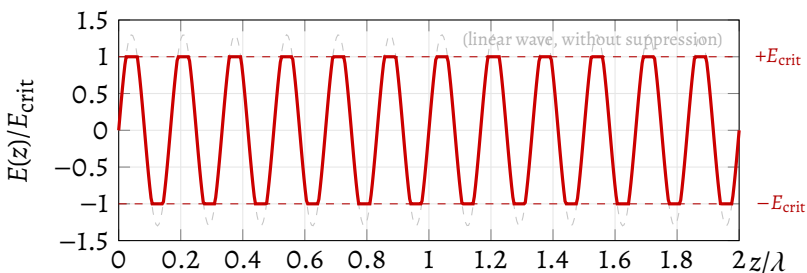
$f < f_{\text{crit}}$ — linear regime



Field never reaches E_{crit} · No semi-particle formed · No electron ejected

(a) Linear regime: $f < f_{\text{crit}}$ (low frequency)

$f > f_{\text{crit}}$ — nonlinear regime activated



Field reaches E_{crit} $\Rightarrow$ nonlinear suppression · Semi-particles formed at the peaks · Ejection if the energy exceeds ϕ

(b) Nonlinear regime: $f > f_{\text{crit}}$ (high frequency)

Figure 10 – Two regimes of *formation* of semi-particles under the hypothesis of restitutive nonlinearity (cf. Appendix R).
(a) For $f < f_{\text{crit}}$ (for a fixed amplitude V_0), the amplitude remains below E_{crit} (dashed red): linear regime, with no semi-particles and hence no ejection of electrons.
(b) For $f > f_{\text{crit}}$, the field would reach amplitudes above E_{crit} (gray ghost), but the nonlinear suppression limits the peaks (solid red curve), forming localized excitations (semi-particles); each one ejects an electron if its energy exceeds the work function of the metal.

